Name: ______________________

Class: ______________________

Date: ______________________

Mr. Rawson • GCSE Physics

The 100 m Dash

Scalars and vectors, distance and displacement, speed and velocity.

Watch these first — tap the blue links.

Bolt's 9.58 s world record, Berlin 2009 where the numbers below come from
BBC — humanoid robots race in Beijing the games themselves
A robot beats Bolt's 100 m record check the time it claims against section 1
Every record the robots broke the 1500 m and the high jump went too

You need these from the first question onwards.

Speed $= \dfrac{\text{distance}}{\text{time}}$ a **scalar** **Velocity** $= \dfrac{\text{displacement}}{\text{time}}$ a **vector**

The exam writes this as $s = v\,t$, so $v = \frac{s}{t}$ and $t = \frac{s}{v}$. *Equation first, then substitute, then a unit.*

1 A robot beat Usain Bolt

On **22 August 2026**, at the World Humanoid Robot Games in Beijing, a robot called **Tiangong Ultra** ran 100 m in **9.39 s**. Usain Bolt's world record, set in Berlin in 2009, is **9.58 s**.

1) **Calculate** the average speed of each, to 3 significant figures.

Tiangong Ultra

Usain Bolt

2) **State** which was faster, and by how much.

And that was only the warm-up. Over the five days of the games the robot's own record fell twice more — **9.39 s**, then **8.86 s**, then **8.64 s** in the closing final. Its top speed was **14.5 m/s**. Bolt's fastest ever measured speed was **12.32 m/s**.

2 Distance and displacement

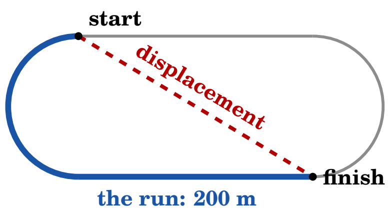
Distance: How far something travels. A **scalar**.

Displacement: How far it ends up from where it started, *and in which direction*. A **vector**.

The 100 m is run in a **straight line**, so the two are the same. Longer races are not run in a straight line — and that is where they come apart.

Race	Distance travelled	Displacement
100 m — straight		
400 m — one full lap		

3) **Explain** why the 400 m runner's displacement is not 400 m.



4) **Explain** why the 200 m runner's **distance travelled** and **displacement** are not the same. The diagram shows the race: the blue line is the route run, the dashed red line is the displacement.
*The 200 m is exactly **half a lap** — one bend and one straight. You are not expected to work out a number: just say why the two differ, and which is larger.*

3 Scalars and vectors

Scalar: Size only.

Vector: Size and direction.

5) **Circle** **S** for scalar or **V** for vector.

9.58 s S / V   100 m S / V   10.4 m/s east S / V   94 kg S / V

4 Using $s = vt$

6) **Calculate** how far Bolt travels in 4.0 s at his average speed.

7) **Explain** why a 400 m runner can have a large *speed* but a **velocity of zero**.

5 Bolt was not going the same speed all race

Every 10 m of the record run, measured by laser. **Time** is for that 10 m section alone. **Acceleration** is how quickly his speed was changing — a *negative* value means he was slowing down.

Distance (m)	10	20	30	40	50	60	70	80	90	100
Time (s)	1.46	0.99	0.90	0.85	0.83	0.82	0.81	0.82	0.82	0.81
Velocity (m/s)	9.12	10.60	11.46	11.95	12.25	12.32	12.32	12.24	12.26	12.27
Accel. (m/s^2)	2.60	0.70	0.60	0.60	0.35	−0.35	−0.15	−0.08	−0.15	−0.35

8) **Identify** the distance at which Bolt **stopped speeding up**, and say how the table tells you.

9) **Suggest** why the first 10 m takes so much longer than any other.

6 The robot never gets tired

Tiangong Ultra won the longer races too. Work out the average speed for each row.

Race	Robot's time	Robot's speed	Human record	Human speed
100 m	9.39 s		9.58 s	
400 m	38.15 s		43.03 s	
1500 m	141.64 s		206.00 s	

10) **Describe** what happens to the *human* speeds as the race gets longer, and what happens to the *robot's*.

11) **Suggest** why the robot's speed barely changes.

Bolt's fastest measured speed was 12.32 m/s*, at the 52 m mark — Štuhec et al. (2023), laser-measured.*

The data behind the record

Bolt's 9.58 s in Berlin, 2009 — the fastest 100 m ever run. Every number here was measured by laser, 100 times a second. *Not for marking. Just look at how precise it gets.*

The race in three phases

	Speeding up	Top speed	Slowing down
Lasted	4.44 s	4.19 s	0.47 s
…which is	48.8% of the race	48.0%	5.2%
Covered	42.80 m	51.28 m	5.56 m
From …to	0 – 42.8 m	42.3 – 94.1 m	94.1 – 99.6 m

He spends almost half the race still accelerating — and he is slowing down for the last 5 m of it.

How long it took to reach each speed

% of top speed	40%	50%	60%	70%	80%	90%	**100%**
Velocity (m/s)	4.95	6.18	7.43	8.65	9.88	11.10	**12.33**
Reached at (m)	1.12	2.37	4.50	8.43	12.72	23.94	**52.51**
After (s)	0.27	0.49	0.80	1.29	1.75	2.82	**5.24**

He is at 90% of top speed by 24 m — but needs another 29 m to find the last 10%.

Bolt against the seven other finalists

	Bolt	The rest of the final (mean)
Time (s)	**9.58**	9.91
Average velocity (m/s)	**10.44**	10.09
Steps taken	**41** (21 right, 20 left)	44.91
Step length (m)	**2.449**	2.23
Step frequency (Hz)	4.472	4.53

Look at the last two rows. Bolt's legs turn over *slightly slower* than everyone else's — he wins because each step is 22 cm longer, so he needs four fewer of them.

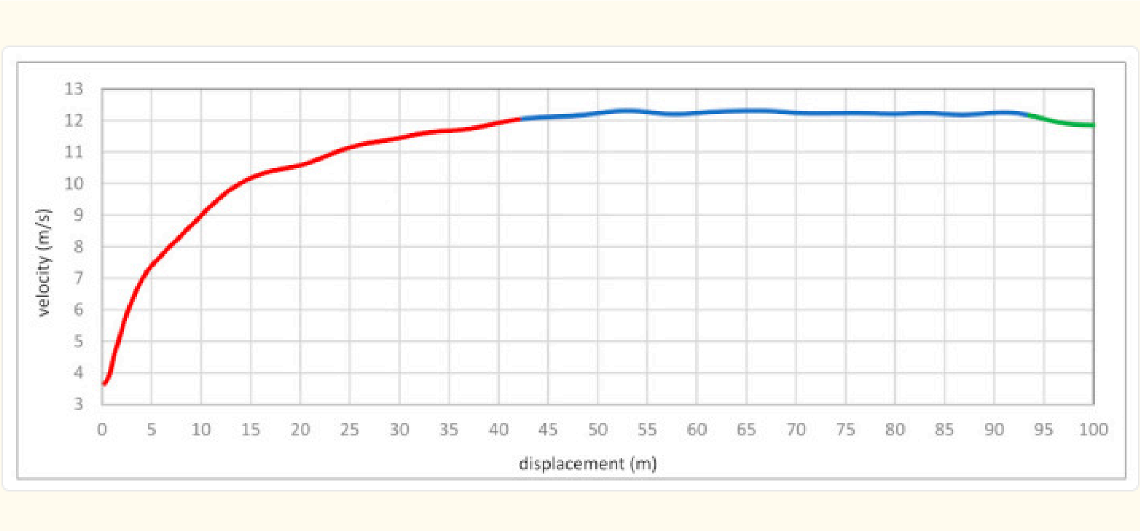
All data from Štuhec, Planjšek, Čoh & Mackala (2023), *Multicomponent Velocity Measurement for Linear Sprinting: Usain Bolt's 100 m World-Record Analysis*, *Bioengineering* **10**(11), 1254. `doi.org/10.3390/bioengineering10111254`. Open access, CC BY 4.0. Laser measurement (LAVEG). Bolt's step data is from Štuhec's software v3.0; the finalists' from the IAAF.

Appendix — the graphs from the study

These are the actual figures from the research paper. You are not asked to do anything with them — they are here because they are worth looking at.

The whole race, as a velocity–displacement graph

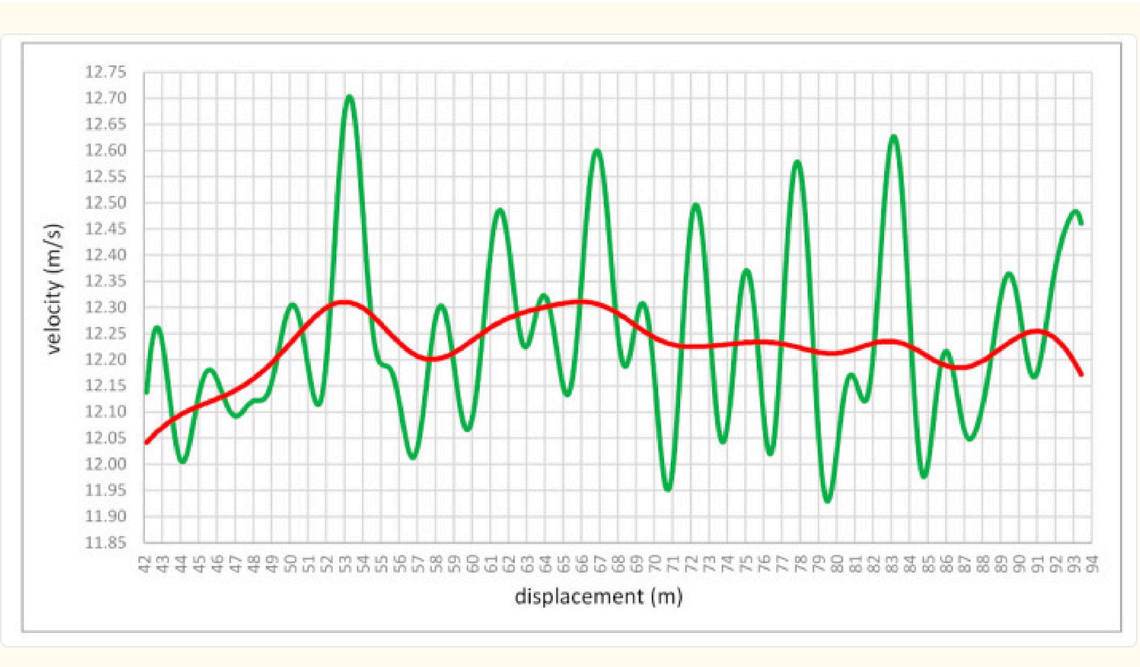
Red = speeding up • blue = top speed (within 2% of maximum) • green = slowing down.



The curve is steep to about 20 m, flattens into a long plateau, and tips down only in the last few metres. **This is the graph you will draw for yourself in Lesson 3.**

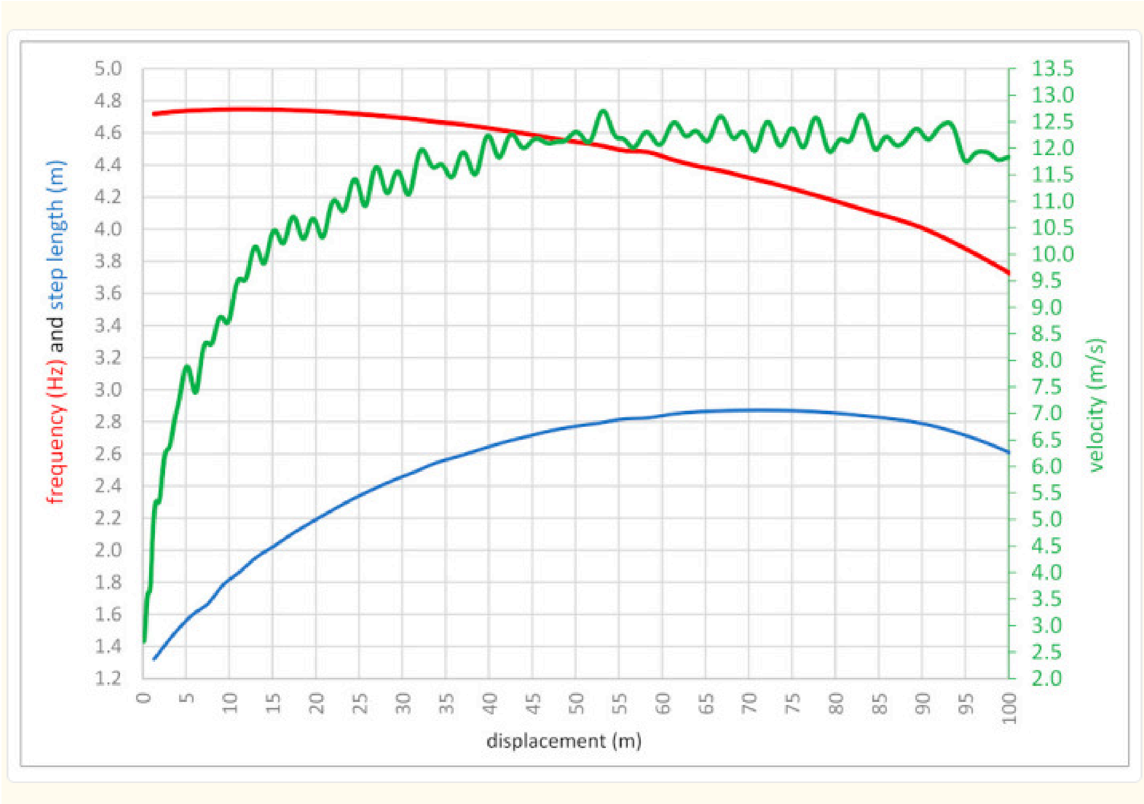
Even the "constant" bit is not constant

Green = raw laser data • red = the smoothed average. Note the scale: this is 42–94 m, and the whole vertical axis spans less than 1 m/s.



Why he wins: step length against step frequency

Red = step frequency (Hz) • blue = step length (m) • green = velocity (m/s).



His step *frequency* falls away for the whole second half of the race. His step *length* keeps rising until about 70 m. That trade is the race.
